INCH-POUND

MIL-P-70528 (AR)

29 OCTOBER 1990

MILITARY SPECIFICATION

PROJECTILE, 105MM, HE, RA, XM913/XM927 DELAY ASSEMBLY PARTS FOR

This specification is approved for use within the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification contains requirements and quality assurance provisions for the fabrication and packaging of delay assembly parts for the 105MM Rocket Assisted Projectiles, XM913 and XM927 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For

STANDARDS

MILITARY

MIL-STD-1169 - Packaging, Packing and Marking for Interplant Shipment of Inert Ammunition

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 1315

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(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from: Military Specifications and Standards, Bldg 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS (See 6.5)

US ARMY RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

PRODUCT AND PACKAGING DRAWINGS

12929177	- Housing, Delay
9381149	- Closure, Delay

(Copies of handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the test of this document and the references cited herein the text of this document takes precedence. Nothing in the document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Parts. The parts shall comply with all requirements specified on drawings listed in 2.1 and associated drawings and with all requirements specified in applicable specifications and standards.

3.3 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of the first article sample by the contractor shall be as specified in the contract.

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3.4 Workmanship. All parts shall be fabricated and finished in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, unblended radii, tears, holes, porosity, dirt, and grease. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. All required markings shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified herein or in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facility suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspection are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, in an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defect material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First Article Inspection
b. Quality Conformance Inspection

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4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated:

Part description	Drawings	Quantity
Housing, Delay	12929177	10
Cap, Selector	9381139	10
Washer, Delay	9381145	10
Gasket	9381148	10
Closure, Delay	9381149	20 (Consecutively produced from each cavity of each mold)

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. See MIL-A-48078.

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TABLE I. First article inspection

CLASSIFICATION OF CHARACTERISTICS

MIL-P-70528 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
	Delay Assembly Parts	OF 1 2	See below
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Housing, Delay (Dwg. 12929177) Examination for defects	10	3.2	4.4.2.1
	Washer, Delay (Dwg. 9381145) Examination for defects	10	3.2	4.4.2.2
	Closure, Delay (Dwg. 9381149) Examination for defects	20/cavity	3.2	4.4.2.3
	Gasket (Dwg. 9381148) Examination for defects	10	3.2	4.4.2.4
	Cap, Selector (Dwg. 9381139) Examination for defects	10	3.2	4.4.2.5

NOTES:

AMSMC Form 1570b, 1 Jul 89 Replaces 1570, 1 Feb 85, which may not be used.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examination and tests.

a. Classification of Characteristics. Quality conformance examinations and tests are specified in the following Classification of Characteristics paragraphs. The contractor's quality program or detailed inspection system shall provide assurance of compliance of all characteristics with the applicable drawing and specification requirements utilizing as a minimum the conformance criteria specified herein. Attributes sampling inspection shall be conducted in accordance with Table II below, using the inspection levels cited in the Classification of Characteristics paragraphs.

TABLE II 0 - 1 ATTRIBUTE PLANS

Lot Size	INSPECTION LEVELS					
	I	II	III	IV	V	VI
2 to 8	*	*	*	*	5	3
8 to 15	*	*	*	13	5	3
15 to 25	*	*	*	13	5	3
26 to 50	*	*	32	13	5	3
51 to 90	*	*	32	13	12	4
91 to 150	*	125	32	13	12	5
151 to 280	*	125	32	30	14	6
281 to 500	*	125	32	30	17	7
501 to 1200	*	125	74	35	20	9
1201 to 3200	1250	125	74	43	24	10
3201 to 10000	1250	125	87	50	30	10
10001 to 35000	1250	296	109	61	36	10

Numbers under inspection levels indicate sample size; asterisks indicate one hundred percent inspection. Accept on zero and reject on one or more for all inspection levels.

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b. Alternative inspection provisions. Alternative inspection procedures, methods or equipment, such as statistical process control, tool control, other types of sampling procedures, etc., may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions specified herein. Prior to applying such alternative procedures, methods or equipment, the contractor shall describe them in a written proposal submitted to the Government for evaluation and approval (see 6.3). When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the specified quality assurance provisions herein. In cases of dispute as to whether the contractor's proposed alternative provides equal assurance, the provisions of this specification shall apply. All approved alternative inspection provisions shall be specifically incorporated into the contractor's quality program plan or detailed inspection system, as applicable.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Housing, Delay	1 OF 3	12929177
			NEXT HIGHER ASSEMBLY 12926809

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Delay column hole does not break through counterbore at tube	Level III	3.2	Visual
102	Diameter of column hole	Level III	3.2	Gage
103	Depth of column hole	Level III	3.2	Gage
104	Position of column hole to tube	Level III	3.2	Gage
105	Position of washer seat	Level III	3.2	Gage
106	Diameter of washer seat	Level III	3.2	Gage
107	Runout of surface to be knurled	Level III	3.2	Gage
108	Diameter of surface prior to knurl	Level III	3.2	Gage
109	Knurl missing	Level III	3.2	Visual
110	Diameter of aft O-ring groove	Level III	3.2	Gage
111	Runout of aft -Oring groove	Level III	3.2	Gage
112	Minimum effective thread	Level III	3.2	Gage
113	Major thread diameter	Level III	3.2	Gage
114	Runout of threads	Level III	3.2	Gage
115	Pitch diameter of threads	Level III	3.2	Gage
116	Location of knurled surface	Level III	3.2	Gage
117	Location of O-ring groove on taper	Level III	3.2	Gage
118	Depth of O-ring groove on taper	Level III	3.2	Gage
119	Location of large basic diameter on taper	Level III	3.2	Gage
120	Circular runout of large basic diameter	Level III	3.2	Gage

NOTES:

Replaces 1570, 1 Feb 85, which may not be used.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-P-70528 (AR)

PARAGRAPH	TITLE	SHEET 2 OF 3	DRAWING NUMBER 12929177
4.4.2.1	Housing, Delay		NEXT HIGHER ASSEMBLY 12926809

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
121	Location of small basic diameter on taper	Level III	3.2	Gage
122	Circular runout of small basic diameter	Level III	3.2	Gage
123	Outside diameter of tube	Level III	3.2	Gage
124	Radius on forward end of transfer hole missing	Level III	3.2	Visual
124	Runout of tube	Level III	3.2	Gage
125	Protective finish missing or incomplete	Level III	3.2	Visual
126	Position of washer seat	Level III	3.2	Gage
127	Diameter of washer seat	Level III	3.2	Gage
Minor				
201	Width of aft O-ring groove	Level V	3.2	Gage
202	Width of O-ring groove on taper	Level V	3.2	Gage
203	Corner radius of O-ring groove on taper (2 places)	Level V	3.2	Gage
204	Surface finish in O-ring groove on taper	Level V	3.2	Visual
205	Width of raised surface at end of threads	Level V	3.2	Gage

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Housing, Delay	3 OF 3	12929177
			NEXT HIGHER ASSEMBLY: 12926809

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
206	Diameter of raised surface	Level V	3.2	Gage
207	Diameter aft end (threaded end)	Level V	3.2	Gage
208	Distance from aft end to start of threads	Level V	3.2	Gage
209	Depth of washer seat	Level V	3.2	Gage
210	Distance aft end to shoulder at tube	Level V	3.2	Gage
211	Distance aft end to bevel below knurl	Level V	3.2	Gage
212	Width of bevel	Level V	3.2	Gage
213	Overall length	Level V	3.2	Gage
214	Width of knurled surface (includes bevel)	Level V	3.2	Gage
215	Surface finish in side column	Level V	3.2	Visual
216	Evidence of poor workmanship	Level V	3.4	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-P-70528 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Washer, Delay	1 OF 1	9381145

NEXT HIGHER ASSEMBLY: 12926809

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Outside diameter	Level III	3.2	Gage
Minor				
201	Inside diameter	Level V	3.2	Gage
202	Protective finish missing	Level V	3.2	Visual
203	Thickness	Level V	3.2	Gage
204	Evidence of poor workmanship	Level V	3.4	Visual

NOTES:

AMSMC Form 1570b, 1 Jul 89 — Replaces 1570, 1 Feb 85, which may not be used.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-P-70528 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	Closure, Delay	1 OF 1	9381149
			NEXT HIGHER ASSEMBLY: 12926809

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Diameter of base	4.4.3.1/0-1	3.2	Gage
102	Length of stem	4.4.3.1/0-1	3.2	Gage
Minor				
201	Overal length	4.4.3.1/0-1	3.2	Gage
202	Evidence of poor workmanship	4.4.3.1/0-1	3.4	Visual

NOTES:

AMSMC Form 1570b, 1 Jul 89 Replaces 1570, 1 Feb 85, which may not be used.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-P-70528 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.4	Gasket	1 OF 1	9381148
			NEXT HIGHER ASSEMBLY: 12926809

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major	None defined			
Minor				
201	Diameter	Level V	3.2	Gage
202	Thickness	Level V	3.2	Gage
203	Evidence of poor workmanship	Level V	3.4	Visual

NOTES:

AMSMC Form 1570b, 1 Jul 89 Replaces 1570, 1 Feb 85, which may not be used.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.5	Cap, Selector	1 OF 1	9381139
			NEXT HIGHER ASSEMBLY: 12826809

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Knurl missing	Level III	3.2	Visual
102	Major inside diameter	Level III	3.2	Gage
103	Minor diameter of threads	Level III	3.2	Gage
104	Pitch diameter of threads	Level III	3.2	Gage
105	Position of threads	Level III	3.2	Gage
106	Protective finish missing or incomplete	Level III	3.2	Visual
Minor				
201	Diameter before knurling	Level V	3.2	Gage
202	Diameter of gasket seat	Level V	3.2	Gage
203	Depth of gasket seat	Level V	3.2	Gage
204	Length to top of thread	Level V	3.2	Gage
205	Overal depth	Level V	3.2	Gage
206	Overal length	Level V	3.2	Gage
207	Marking missing or incomplete	Level V	3.2	Visual
208	Evidence of poor workmanship	Level V	3.4	Visual

NOTES:

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4.4.3 Testing. Testing is described in the First Article and Quality Conformance Inspection tables.

4.4.3.1 Molded components. Subsequent to acceptance of first article and after production of 10,000 pieces per cavity per mold, 20 pieces from each cavity of each mold shall be consecutively produced and inspected for all drawing and specification requirements and verified by the Government inspector. If any part fails to meet all drawing or specification requirements, production shall be stopped and the mold corrected, and all parts produced since the last accepted sample shall be screened for all applicable requirements in accordance with Table II, inspection level IV. During production, the molding procedures shall be monitored for uniformity of molding practices. Suspected deviation in mold product shall be examined to determine if corrective action or mold repair is required. If during production correction or repair is required, 20 consecutively produced units from each affected mold cavity shall be inspected and shall comply with all drawing and specification requirements.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the 'Paragraph Reference/Inspection Method' column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and paragraph 6 herein.

4.5 Methods of inspection.

4.5.1 Materials, components and processes. Compliance with all requirements of Section 3 of this specification shall be ascertained by current and continuing examination of inspection and test data to determine that all components (parts, subassemblies, and materials) have been inspected and tested and found to comply with their respective drawing and specification requirements, and that all specified manufacturing processes have been followed.

4.5.1.1 Visual and mechanical inspections. The visual, manual and mechanical inspections and tests shall verify compliance with the requirements of Sections 3 and 5 of this specification in accordance with 4.4.2 herein and Table I.

5. PACKAGING

5.1 packaging, packing and marking. Packaging, packing and marking shall be in accordance with MIL-STD-1169.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The components covered by this specification are intended for use on the 105MM, XM913 and XM927, HE, RA Projectile.

6.2 Acquisition requirements. See MIL-A-48078 (AR).

6.3 Submission of inspection procedures and equipment designs for approval. See MIL-A-48078 and MIL-STD-45662. Submit equipment designs as required to: Commander, ARDEC, ATTN: AMSMC-QAR-I(D), Picatinny Arsenal, New Jersey 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract. Unless otherwise specified, data item DI-R-1714 will apply.

6.4 Data requirements. Data cards shall be prepared for each lot in accordance with the information specified in MIL-STD-1167.

6.5 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Command (AMCCOM) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities in now under the cognizance of AMCCOM.

6.6 Inspection lots. It is noted that the size of inspection lots of components, assemblies or items of delivery may differ from the actual quantities contractually scheduled for delivery. However, in order to facilitate scheduling of tests or deliveries, inspection lots of items of delivery may be equivalent to such contract quantities provided all of the lot formation criteria and sampling provisions of this specification are maintained.

6.7 Visual examination qualifications. When compliance with the applicable requirement is in doubt as a result of visual examination, the characteristic may be measured or gaged to determine acceptability.

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6.8 Subject term (key word) listing.

Delay parts
Molded components
Selector cap

Custodian:
Army - AR

Preparing activity:
Army - AR

(Project 1315-AC12)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-P-70528(AR)	29 October 1990

3. DOCUMENT TITLE

PROJECTILE, 105MM, HE, RA, XM913/XM927 DELAY ASSEMBLY PARTS FOR

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
US Army ARDEC Standardization & Specification Office	(1) Commercial (201) 724-6671	(2) AUTOVON 880-6671
c. ADDRESS (Include Zip Code) ATTN: SMCAR-BAC-S Picatinny Arsenal, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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